TS – VI (SS) – 11.13 – 0881

B.Tech. Degree VI Semester Special Supplementary Examination November 2013

ME 604 HEAT AND MASS TRANSFER

(2006 Scheme)

Time : 3 Hours | Maximum Marks : 100

(Use of heat and mass transfer data book is permitted)

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Define the terms "fin efficiency' and 'fin effectiveness'.

(b) What is a lumped system? In what medium is the lumped system assumption more likely to be applicable: in water or in air? Why?

(c) What is Reynolds analogy? What are its limitations?

(d) Differentiate between film and drop-wise condensation. Which is a more effective mechanism of heat transfer?

(e) What is a black body? List the key attributes of a black body.

(f) What is a reradiating surface? What simplification does a reradiating surface offer in the radiation analysis?

(g) What are the common causes of fouling in a heat exchanger? How do the fluid velocity and temperature effect fouling?

(h) How is the NTU of a heat exchanger defined? What does it represent? Is a heat exchanger with a very large NTU necessarily a good one to buy?

PART B

(4 × 15 = 60)

II. Steam at 320°C flows in a stainless steel pipe (k = 15 W/m.°C) whose inner and outer diameters are 5cm and 5.5cm, respectively. The pipe is covered with 3cm-thick glass wool insulation (k = 0.038 W/m. °C). Heat is lost to the surroundings at 5°C by natural convection with a convection coefficient of 15 $W/m^2.°C$. Taking the heat transfer coefficient inside the pipe to be 80 $W/m^2.°C$, determine the rate of heat loss from the steam per unit length of the pipe. Also determine the temperature drops across the pipe shell and the insulation. (15)

OR

III. Derive an expression for the temperature distribution along a fin that is insulated at the tip. Also obtain an expression for the rate of heat transfer from the fin and a relationship for the fin efficiency. (15)

IV. Water is to be heated from 10°C to 80°C as it flows through a 2cm internal diameter, 13m long tube. The tube is equipped with an electric resistance heater, which provides uniform heating throughout the surface of the tube. The outer surface of the heater is well insulated, so that in steady operation all the heat generated in the heater is transferred to the water in the tube. If the system is to provide hot water at a rate of 5 litres/minute, determine the power rating of the resistance heater. Also, estimate the inner surface temperature of the pipe at the exit. (15)

OR

V. (a) What is boiling? What mechanisms are responsible for the very high heat transfer coefficients in nucleate boiling? (4)

(b) Draw the boiling curve and identify the different boiling regimes. Also, explain the characteristics of each regime. (11)

VI. (a) What are the radiation surface and space resistances? How are they expressed? (5)

(b) A furnace is of cylindrical shape with radius, r = 2m and height, h = 2m. The base, top and side surfaces of the furnace are all black and are maintained at uniform temperatures of 500K, 700K and 1400K respectively. Determine the net rate of radiation heat transfer to or from the top surface during steady operation. (10)

OR

VII. (a) What is a radiation shield? Why is it used? (5)

(b) Two very large plates are maintained at uniform temperatures of T_1 = 1000K and T_2 = 800K and have emissivities of $\varepsilon_1 = \varepsilon_2 = 0.5$ respectively. It is desired to reduce the net rate of radiation heat transfer between the two plates to $1/5^{th}$ by placing thin aluminium sheets with an emissivity of 0.1 on both sides between the plates. Determine the number of sheets that need to be inserted. (10)

VIII. Hot oil (c_p = 2200 J/Kg °C) is to be cooled by water (c_p = 4180 J/Kg °C) in a 2-shell-passes and 12-tube-passes heat exchanger. The tubes are thin-walled and are made of copper with a diameter of 1.8cm. The length of each tube pass in the heat exchanger is 3m, and the overall heat transfer coefficient is 340 W/m^2 °C. Water flows through the tubes at a total rate of 0.1 Kg/s, and the oil through the shell at a rate of 0.2 Kg/s. The water enters at 18°C and oil enters at 160°C. Determine the rate of heat transfer in the heat exchanger and the outlet temperatures of water and oil. (15)

OR

IX. (a) Derive an expression for the effectiveness of a parallel-flow heat exchanger. (10)

(b) Consider an oil-to-oil double pipe heat exchanger whose flow arrangement is not known. The temperature measurements indicate that the cold fluid enters at 20°C and leaves at 55°C, while the hot fluid enters at 80°C and leaves at 45°C. Do you think this is a parallel-flow or counter-flow heat exchanger? Why? Assuming the mass flow rates of both fluids to be identical, determine the effectiveness of this heat exchanger. (5)

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